Operating Instructions

Diesel engine-generator set
mtu 10V1600 DS500 / DG10V1600A1I
mtu 10V1600 DS540 / DG10V1600A2I
mtu 12V1600 DS650 / DG12V1600A1I
mtu 12V1600 DS715 / DG12V1600A2I
Application groups 3B, 3D, 3E and 3F

Without engine-generator set controller

SS180184/00E

A Rolls-Royce solution

System designation	Engine model	Application group
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G10F	3B, continuous service, variable load, ICXN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G20F	3B, continuous service, variable load, ICXN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G10F	3B, continuous service, variable load, ICXN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G20F	3B, continuous service, variable load, ICXN
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G70F	3D, emergency standby power, IFN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G80F	3D, emergency standby power, IFN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G70F	3D, emergency standby power, IFN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G80F	3D, emergency standby power, IFN
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G10F	3E, emergency service, ICXN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G20F	3E, emergency service, ICXN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G10F	3E, emergency service, ICXN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G20F	3E, emergency service, ICXN
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G10F	3F, heavy duty for DCP, unrestricted, ICXN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G20F	3F, heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G10F	3F, heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G20F	3F, heavy duty for DCP, unrestricted, ICXN

Table 1: Applicability

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1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

Store these Operating Instructions where they can be accessed by all persons entrusted with operating or servicing this product.

Before starting any work, read these Operating Instructions carefully. For safe operation of the product, always follow all safety instructions and work instructions.

Local accident prevention regulations and general safety regulations for the area of application of the product also apply.

These Operating Instructions form part of the product and refer to the latest design version of the product at the time of publication. They are subject to modification to reflect technical enhancements. Actual product may vary from figures shown.

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2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:http://www.mtu-solutions.com

TIM-ID: 0000040530 - 017

2.2 Intended use of engine-generator sets

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:
- As generating set, stationary applications, for on-site power generation (isolated operation), for power generation as mains backup system (standby power), as well as for mains parallel operation
- As generating set to produce electric power at the voltages / frequencies/ temperatures and the possible maximum power specified on the nameplate
- As generating set in closed rooms, which provide the required technical equipment, such as supply and exhaust air opening, exhaust pipe routing through the wall to the outside, and a fuel supply facility
- Within the admissible operating parameters in accordance with the (→ technical sales documents – product data – TVU data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and observance of all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques

The product shall not be operated in explosive atmospheres unless it fulfills the conditions for such use and an approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to use, ensure that the most recent version is available. The latest version can be found at:
- http://www.mtu-solutions.com

Reasonably foreseeable misuse

- The engine-generator set was not designed for outdoor usage.
- The engine-generator set was not designed for mobile applications.
- The engine-generator set must not be operated with fluids and lubricants which are not listed in the specifications.
- The engine-generator set was not designed for the usage as combined heat and power plant or in applications where special requirements apply, e.g. nuclear power plant, oil&gas, fracking.
- When the specified operating hours, i.e. the expected lifetime of the product, is exceeded, the need for maintenance is expected to increase.
- The engine-generator set shall not be operated at other voltages and frequencies than those specified on the nameplate.
- The total rated power consumption of all consumers connected to the engine-generator set shall not exceed the total output of the engine-generator set specified on the nameplate.

Without safety certification according to IBC or Eurocode8, it is not recommended to use engine-generator sets in regions with high seismic activity, with the influence coefficient being defined as 1.5.

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If lower requirements apply, the seismic force can be estimated according to the FEMA 450 guide using the terms and definitions of DIN 4149 and of DIN EN 1998-1 (2010) respectively. If a_gR is lower than 0.39 m/s^2, no proof of the seismic safety is required as per EN1998.

Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:
- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

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